V2 Rail Guard + Prolabs Flex Sheet (0.2mm)

by Aarav Jaisingh, Prolabs Robotics

Build guide for Bangalore FTC teams | goBILDA FTC StarterBot 2026-27 (BIOBUZZ) | 96mm Hogback, 71mm POLLEN

NOT ROBOT TESTED. We have checked this in CAD only. Updated 13 Nov 2026: plate 368 fix, V2 mounting, corrected V2 hole positions, fit check. The 0.2mm sheet is our biggest worry (see Known issues). Please test one first and share what you find so we can improve it together.

Download the files

3D files folder (V2 Shooter 3D files): https://github.com/Aarav-111/FTC-Bangalore-Resources/tree/main/FTC%20precise%20shooting%20mechanism/V2%20Shooter%203D%20files

FTC Shooter V2 Rail Guard.stl
V2 rail guard, print 1
Download

prolabs_plain_sheet_0p2.stl
Prolabs Flex Sheet, print 1
Download

plate_368_cut_5x5.stl
Plate 368 replacement, 40 x 40 x 1.5mm, print 1
Download

plate_368_5x6_trim.stl
Optional 40 x 48mm version with 1 extra row to trim
Download

plate_368_cut_5x5.step / 5x6_trim.step
STEP files for CAD (Onshape, Fusion)
5x5 STEP | 5x6 STEP

V2_shooter_simulator.html
3D simulator. Download, then open in Chrome or Safari. Works offline. Drag to rotate, pinch to zoom, Shift + drag to pan.
Download

Scan for the 3D files

Scan for all shooter docs

More docs in the FTC precise shooting mechanism folder:

BOM & 3D printing Guide.md
Open

FTC Shooting and Intake Mechanism Theory.pdf
Open

V2_flexplate_build_guide.pdf
This guide
Open

1. Open the link ▶ 2. Click the file ▶ 3. Click Download raw file (top right) ▶ Saved

To get everything at once, we open the repo page and click **Code > Download ZIP**. The simulator is a .html file: download it with Download raw file, then double-click it.

How to print (beginner friendly)

1 Open a slicer

Use any free slicer: **Cura, PrusaSlicer, Bambu Studio or OrcaSlicer**.

2 Import the STL

File > Import, or drag the STL onto the bed.

3 Check scale is 100%

No unit conversion. V2 should read about **129 x 149 x 88mm**. The sheet should read **88 x 104 x 0.2mm**. Plate 368 should read **40 x 40 x 1.5mm** (5x6: 40 x 48 x 1.5mm). If it is tiny or huge, the units are wrong.

4 Set the orientation

V2: lay it sideways on its flat side. **Sheet and plate 368:** flat on the bed.

5 Apply the settings

Use the settings table below for each part.

6 Slice and preview

The sheet must show **exactly 1 layer** with all holes visible. V2 should have no support.

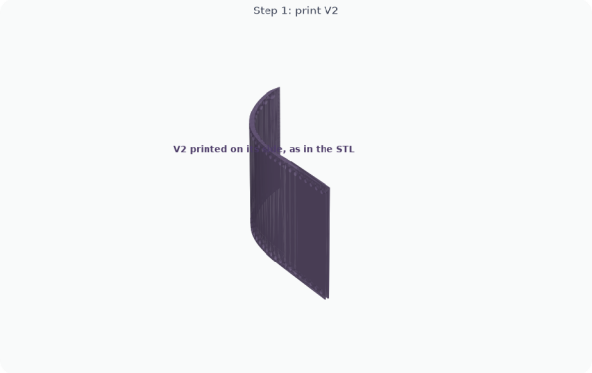
7 Print and remove

Watch the first layer. Peel the sheet off a flexible plate by bending the plate.

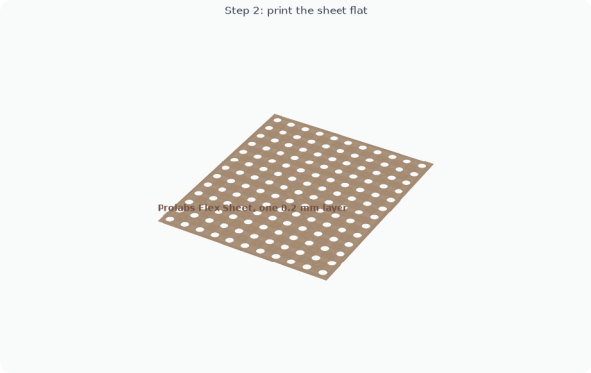
8 Check dimensions

Use calipers. Each size within **+-0.5mm**.

V2 orientation (sideways)



Sheet orientation (flat, one layer)



Setting	V2 rail guard	Prolabs Flex Sheet	Plate 368 (5x5)
Material	PETG or PLA+	PETG	PETG or PLA+
Infill	**100%**	**100%**	**100%**
Orientation	Sideways	Flat	Flat
Supports	None	None	None
Layer height	Normal (0.2mm is fine)	**0.2mm, so it is one layer**	0.2mm (1.5mm thick)
Adhesion	5mm brim	No brim; clean, level bed	No brim
First layer speed	Normal	**Slow, about 15 to 20 mm/s**	Normal
Removal	Normal	Flexible plate, peel gently	Normal

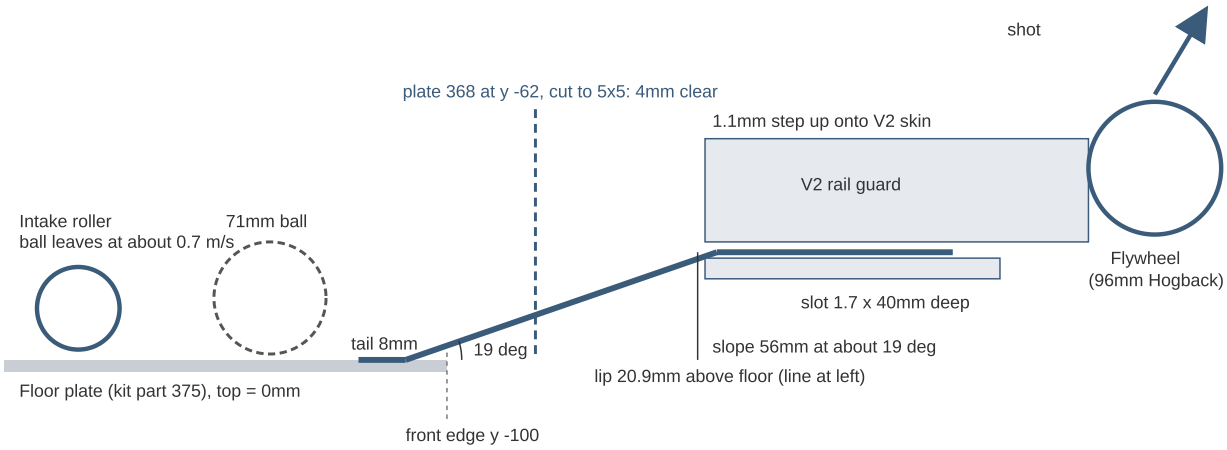
Common problems

Problem	Fix
Sheet will not stick	Clean the bed (soap and water or IPA), re-level, slow the first layer, use a glue stick
Holes closed up	Lower first-layer squish (raise Z offset a little), reduce first-layer flow
Warping (V2)	Keep the brim, raise bed temperature, avoid drafts
Stringing	Dry the PETG, increase retraction, lower nozzle temperature by 5 to 10 C

1. How it shoots (CAD analysis)

This is an analysis from CAD and the STL files. It is not robot tested. Speeds and clearances below are estimates. Test with one ball by hand first.

Side section of the ball path (schematic, not to scale)



Unpowered zone: about 131mm from intake roller to the V2 squeeze. Nothing drives the ball here.

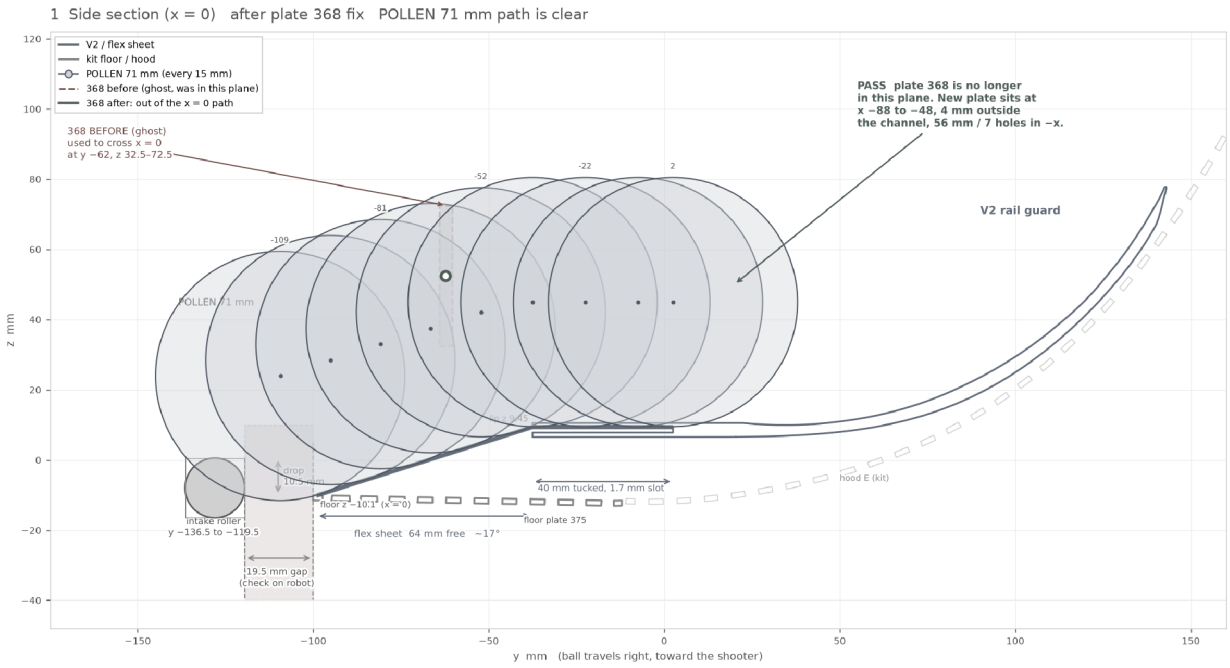
Stage	What happens	Key number
1. Intake roller	Roller throws the ball forward across the floor plate	Needs about 0.7 m/s at exit
2. Floor plate	Ball rolls to the front edge	Front edge y -100
3. Sheet tail and slope	Ball rolls onto the 8mm tail and climbs the 56mm slope	About 19 deg, rise about 18mm
4. Tongue and V2 lip	Sheet tongue sits in the V2 slot. Ball steps up onto the V2 skin	Lip 20.9mm, step 1.1mm
5. V2 rail guard	Rail guard holds the ball against the flywheel	Squeeze peaks at 14.5% on 71mm ball
6. Flywheel and shot	Hogback wheel grabs the ball and launches it along the shooter hood	Not measured

Total unpowered run: about **131mm**, partly uphill. If the ball is slower than about 0.7 m/s it can stop on the slope. A second ball behind it can push it through.

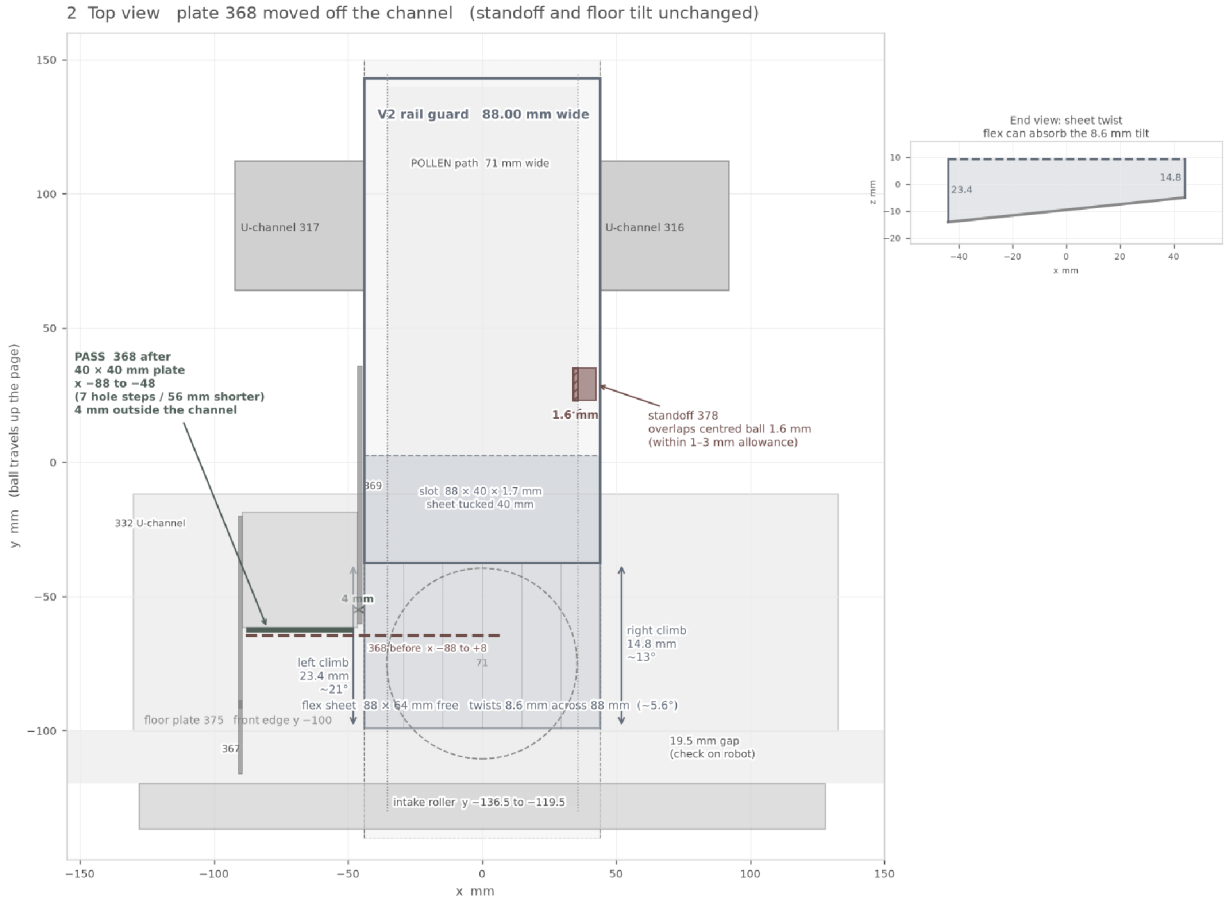
Expected failure points

Where	What can go wrong	What to check
Slope	Ball stalls, too slow from the roller	Roll one ball at match speed
0.2mm sheet	Sags under the ball, bend angle changes	Watch the sheet while rolling
V2 lip	Ball catches on the 1.1mm step	Chamfer the lip, roll by hand
Part 368 (y -62)	Fixed: cut to 5x5 (40 x 40mm). 4mm clear of the ball in CAD	Roll a ball past it
Roller to floor gap	Ball coasts across a 19.5mm gap	Check on the robot
V2 mount	V2 lifts when the ball hits it	At least 2 ties per side, no wobble
Chains	Band pulls the tongue out or tears holes	Tongue stays 40mm in the slot
Squeeze	14.5% is the top of the 10 to 15% target, can jam soft balls	Feel the ball pass under V2

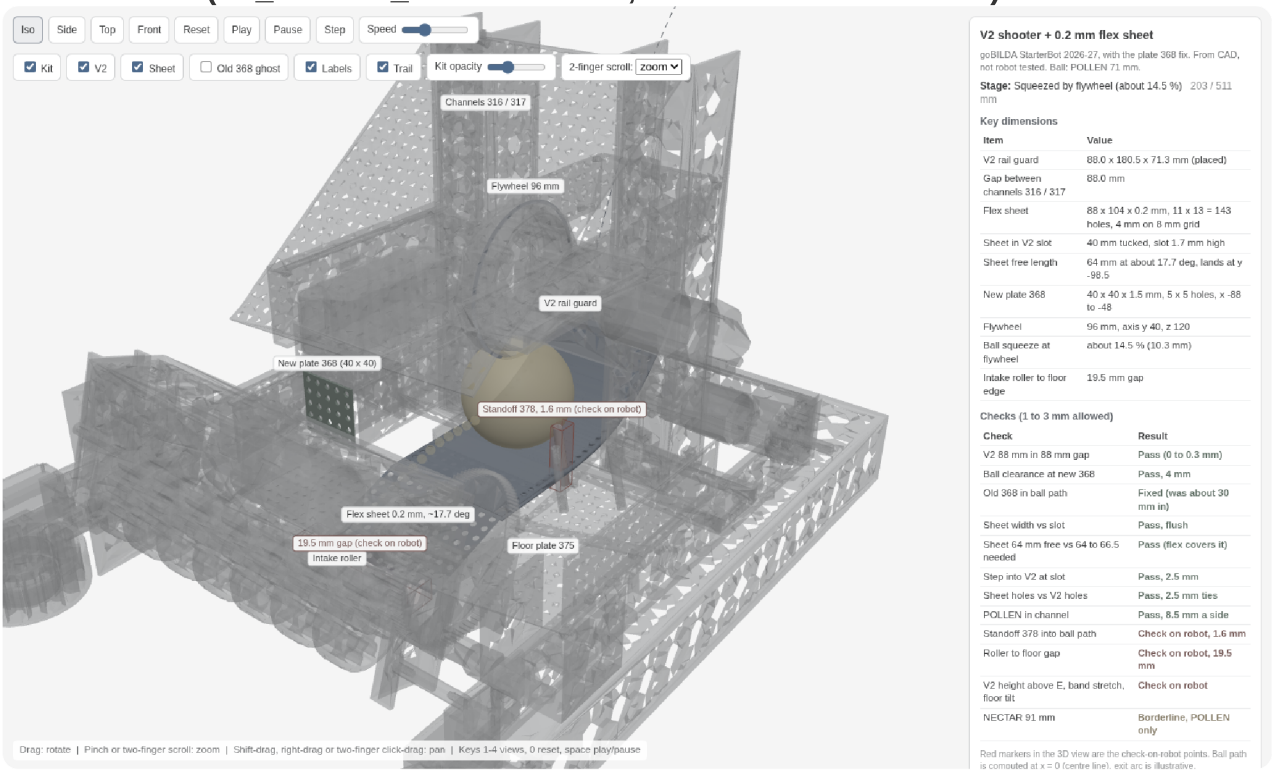
Simulation: side section, POLLEN 71mm path after the plate 368 fix



Simulation: top view, plate 368 cut, standoff 378 and floor tilt



3D simulator (V2_shooter_simulator.html, see Download the files)



V2 shooter + 0.2 mm flex sheet

goBILDA StarterBot 2026-27, with the plate 368 fix. From CAD, not robot tested. Ball: POLLEN 71 mm.

Stage: Squeezed by flywheel (about 14.5 %) 203 / 511 mm

Key dimensions

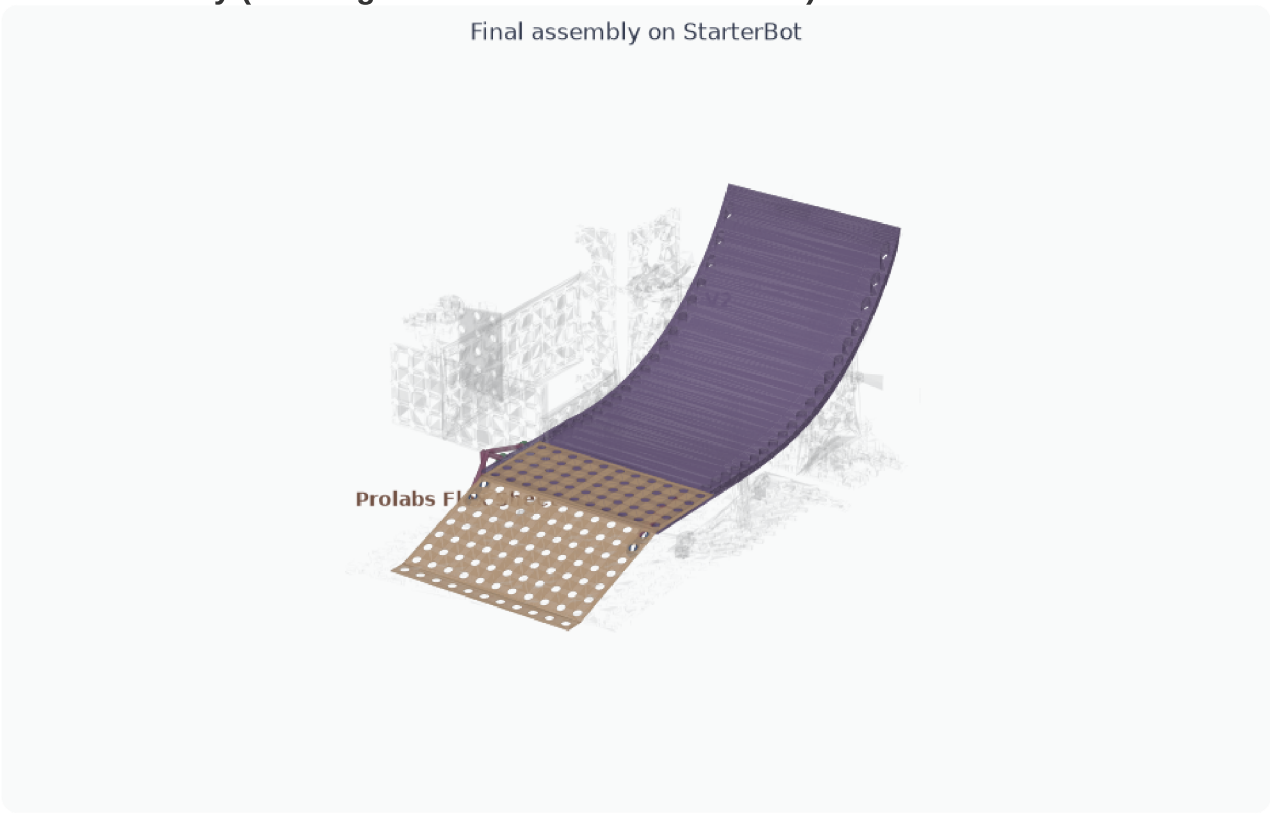
Item	Value
V2 rail guard	88.0 x 180.5 x 71.3 mm (placed)
Gap between channels 316 / 317	88.0 mm
Flex sheet	88 x 104 x 0.2 mm, 11 x 13 = 143 holes, 4 mm on 8 mm grid
Sheet in V2 slot	40 mm tucked, slot 1.7 mm high
Sheet free length	64 mm at about 17.7 deg, lands at y -98.5
New plate 368	40 x 40 x 1.5 mm, 5 x 5 holes, x -88 to -48
Flywheel	96 mm, axis y 40, z 120
Ball squeeze at flywheel	about 14.5 % (10.3 mm)
Intake roller to floor edge	19.5 mm gap

Checks (1 to 3 mm allowed)

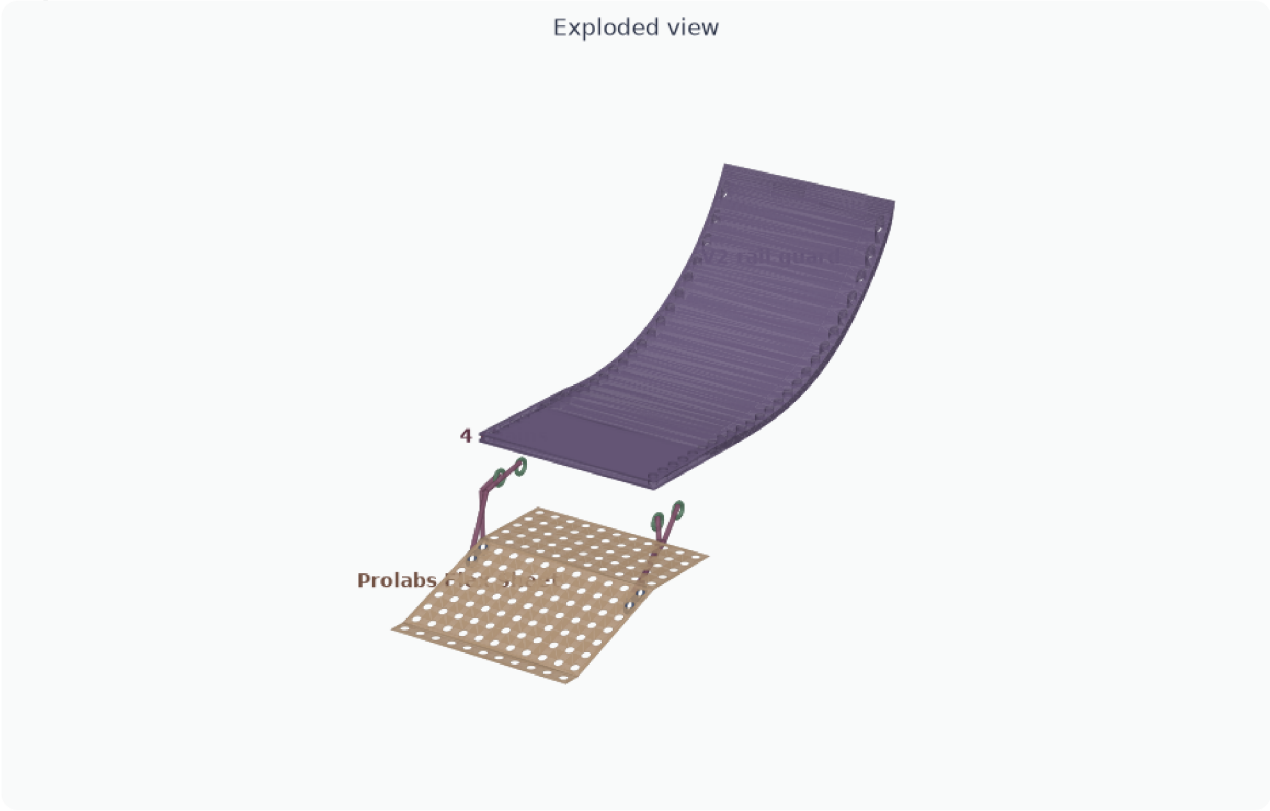
Check	Result
V2 88 mm in 88 mm gap	Pass (0 to 0.3 mm)
Ball clearance at new 368	Pass, 4 mm
Old 368 in ball path	Fixed (was about 30 mm in)
Sheet width vs slot	Pass, flush
Sheet 64 mm free vs 64 to 66.5 needed	Pass (flex covers it)
Step into V2 at slot	Pass, 2.5 mm
Sheet holes vs V2 holes	Pass, 2.5 mm ties
POLLEN in channel	Pass, 8.5 mm a side
Standoff 378 into ball path	Check on robot, 1.6 mm
Roller to floor gap	Check on robot, 19.5 mm
V2 height above E, band stretch, floor tilt	Check on robot
NECTAR 91 mm	Borderline, POLLEN only

Red markers in the 3D view are the check-on-robot points. Ball path is computed at x = 0 (centre line). exit arc is illustrative.

Final assembly (V2 rail guard and Prolabs Flex Sheet)



Exploded view: V2, sheet, 4 chains



2. Plate 368 fix (needed)

Problem: kit plate 368 is the back wall of the left servo box. It sits between plates **367** (left) and **369** (right), under U-channel **332**, about 62mm back from the floor front edge. In the kit it is 12 x 5 holes (96 x 40mm) and its right end reaches about 30mm into the ball channel.

Fix (pick one):

Option	What to do	Result
A. Print (recommended)	Print **plate_368_cut_5x5.stl**: 40 x 40 x 1.5mm, 25 holes of 4mm on an 8mm grid, 4mm edge. 100% infill, flat on the bed.	Fits as is, all holes line up with 367 and 332
B. Cut the kit plate	Cut **56mm (7 hole rows)** off the channel end of kit plate 368. Keep the end at plate 367. File the cut edge smooth.	Same 40 x 40mm part
C. 5x6 trim version	Print **plate_368_5x6_trim.stl**: 40 x 48mm, one extra row added **downward**. Trim it to fit.	Extra row has no matching hole on 367, it is only spare to cut

Fitting: keep the **left edge against plate 367**, same height as the old plate (x -88 to -48, z 32.5 to 72.5 in the kit CAD). Re-use the kit screws. The POLLEN ball then has **4mm of clearance**.

Before and after: plate 368 in the ball path

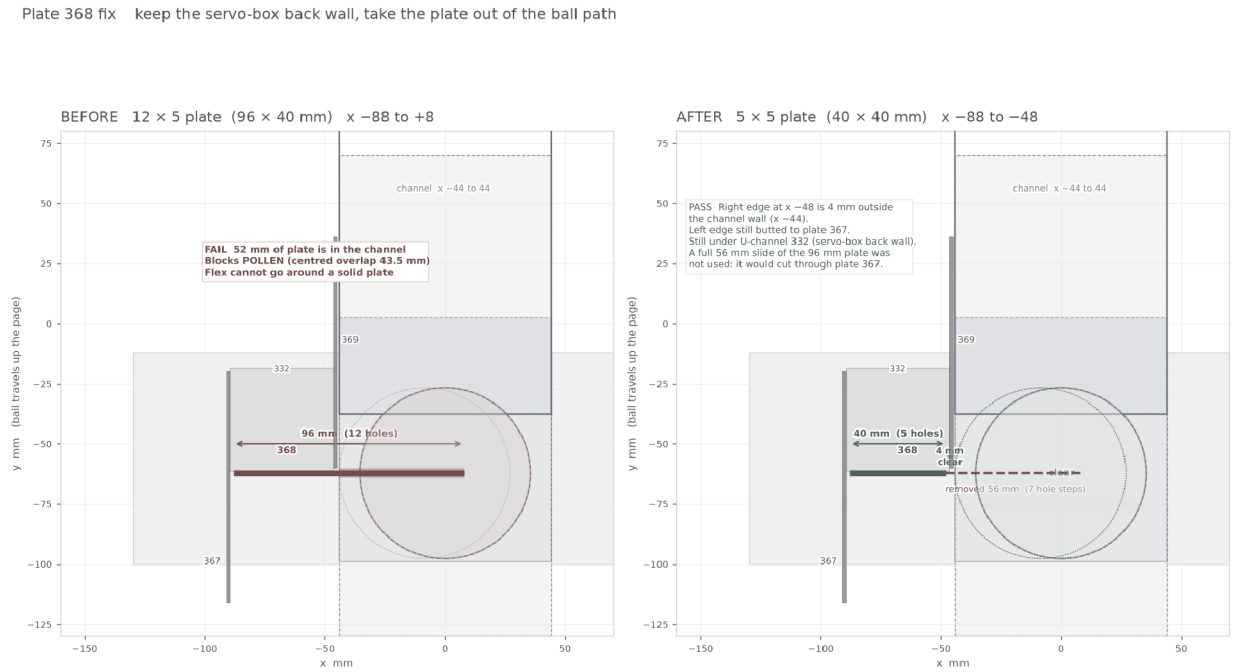


Plate 368 cut to 5x5, and the optional 5x6 trim version

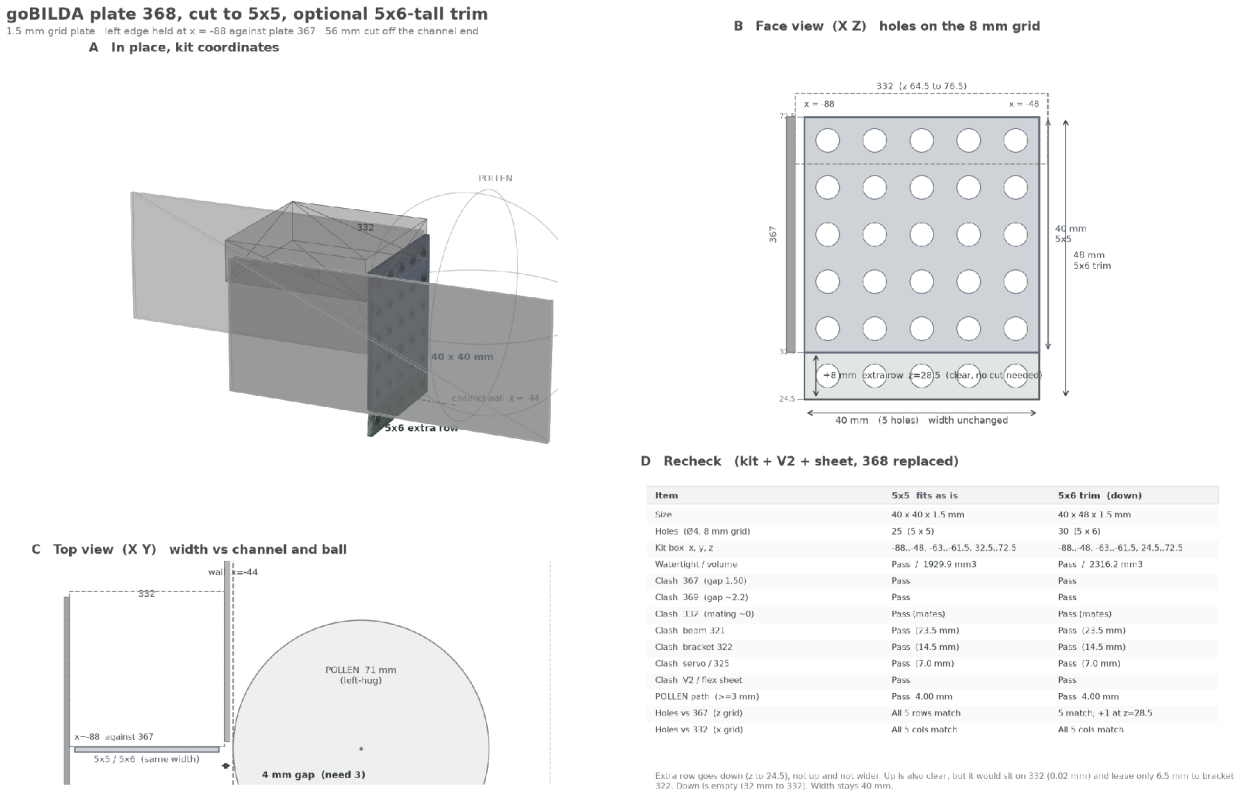


Plate 368 (5x5) check	goBILDA spec	STL	Result
Size	40 x 40mm	40.00 x 40.00mm	Pass
Thickness	1.5mm	1.50mm	Pass
Holes	25 (5x5), 4mm	25, 4.0 to 4.1mm	Pass
Pitch / edge	8mm / 4mm	8.0mm / 4.0mm	Pass
Clash with 367, 369, 332, 321, 322, servo, V2, sheet	none	none	Pass
POLLEN ball clearance	3mm or more	4mm	Pass

3. Dimensions to double-check

Item	Value	Measure with	Tolerance	Checked
Sheet size	88 x 104mm	Calipers	+-0.5mm	STL
Sheet thickness	0.2mm (one layer)	Calipers / micrometer	+-0.05mm	STL
Hole size	4.0mm	Calipers inside jaws	+-0.2mm	STL
Hole pitch	8.0mm, centres at 4 + 8i	Calipers, over 10 holes = 80mm	+-0.5mm over 10	STL
Hole count	11 x 13 = 143	Count	exact	STL
Edge margin	4mm to hole centre, 2mm to hole edge	Calipers	+-0.3mm	STL
Bend 1 location	40mm (5 holes) from tongue end	Ruler	+-1mm	Design
Bend 2 location	8mm (1 hole) from tail end	Ruler	+-1mm	Design
Bend angle	about 19 deg	Protractor	+-3 deg	Design
Sheet weight	about 1.9g PETG	Scale	+-0.3g	STL volume 1473 mm3
V2 overall size	129.35 x 148.57 x 88mm	Calipers	+-0.5mm	STL
V2 volume / weight	55.0 cm3, about 70g PETG	Scale	+-5g	STL
V2 slot	1.7mm high (z 7.75 to 9.45), 40mm deep	Feeler gauge / 0.2mm sheet test fit	+-0.2mm	STL
V2 side holes	4mm, at x +-40	Calipers	+-0.2mm	STL
V2 hole positions under slot	y -32.5, -25.4, -18.3, -11.3, -4.2 (corrected; the old values were the solid parts between holes)	Calipers from front face	+-0.3mm	STL
V2 hole pitch vs sheet	7.08mm vs 8.0mm. Mismatch about 0.9mm per hole. 0.1 to 2.7mm off, all 5 usable with 2.5mm ties	Calipers	-	STL

Lip height above floor plate	20.9mm	Calipers depth rod	+-0.5mm	Earlier kit CAD, not re-checked
Step at lip	1.1mm	Calipers	+-0.2mm	Earlier kit CAD, not re-checked
Floor plate front edge	y -100	-	-	Earlier kit CAD, not re-checked
Plate 368 (new)	40 x 40 x 1.5mm at y -62, left edge on 367	Calipers	+-0.5mm	STL + STEP
Squeeze	14.5% peak on 71mm ball	Feel / gap gauge	-	Earlier analysis, not re-checked
Zip ties	**2.5mm wide** x 100mm. Only 2.5mm ties pass through both 4mm holes	-	-	Checked vs holes
Rubber bands	light, about 1.5mm wide. Must stretch over a **42 to 49mm** span (chain span 58mm right, 61mm left)	Ruler	-	Check on robot

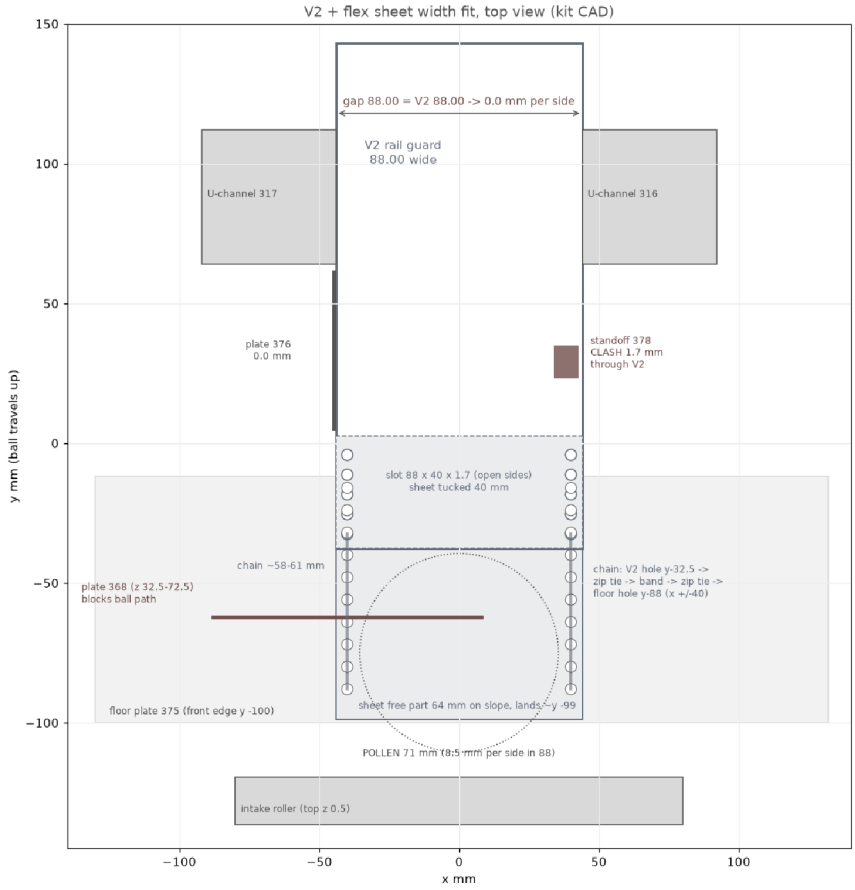
Total sheet length check: tongue 40 + slope 56 + tail 8 = **104mm**.

4. Fit check (1 to 3mm tolerance allowed)

Everything passes in CAD. Four items can only be checked on the robot.

Item	Off by	Result
V2 width vs 88mm gap between channels 316 and 317	0 to 0.3mm	Pass
Standoff 378 into V2 / into ball path	1.7mm / 1.6mm	Pass
Sheet width vs V2 slot / vs floor	0mm / lots of room	Pass
POLLEN ball (71mm) in 88mm channel	8.5mm room per side	Pass
Sheet in 1.7mm slot	1.5mm loose	Pass
Step into V2 at slot	2.5mm	Pass
Sheet length on slope	1 to 2.5mm short	Pass (sheet flexes)
Sheet holes vs V2 holes	0.1 to 2.7mm	Pass with 2.5mm ties
V2 mounting with rubber bands and zip ties	bands cover the 3 to 38mm hole offset to E	Pass
Plate 368 after the cut	4mm outside the channel	Pass
V2 height above hood E	2 to 22mm	Check on robot: tie down so it does not move
Band stretch	42 to 49mm span	Check on robot
Floor tilt (CAD)	about 6 deg, 8.6mm side to side	Check on robot (sheet twists about 5.6 deg to take it up)
Intake roller to floor gap	19.5mm plus 10.5mm drop	Check on robot
NECTAR ball (91mm)	3mm too wide	Borderline. POLLEN only

V2 width fit, top view



5. Parts list

Item	Qty	Notes
V2 rail guard (printed)	1	FTC Shooter V2 Rail Guard.stl
Prolabs Flex Sheet (printed)	1	prolabs_plain_sheet_0p2.stl
Plate 368 replacement (printed, or cut kit plate)	1	plate_368_cut_5x5.stl, 100% infill, flat
Zip ties, 2.5mm wide	**12**	8 for the 4 sheet chains + at least 4 to mount V2 (2 per side)
Rubber bands	**6**	4 for the sheet chains + 2 to mount V2 (1 per side), all the same size
Super glue (thin CA)	1 tube	Used in step 6: zip tie locks (8) and the tongue top edge in the V2 slot. Plus CA primer/activator for PETG
File or knife	-	Chamfer V2 lip
Calipers, 71mm ball	-	Hand checks

6. Print settings

V2 rail guard

- PETG
- On its side
- No supports
- **100% infill**
- 5mm brim

Prolabs Flex Sheet

- PETG, printed flat
- **100% infill**, one 0.2mm layer
- Level the bed carefully
- Slow first layer
- Flexible build plate for removal

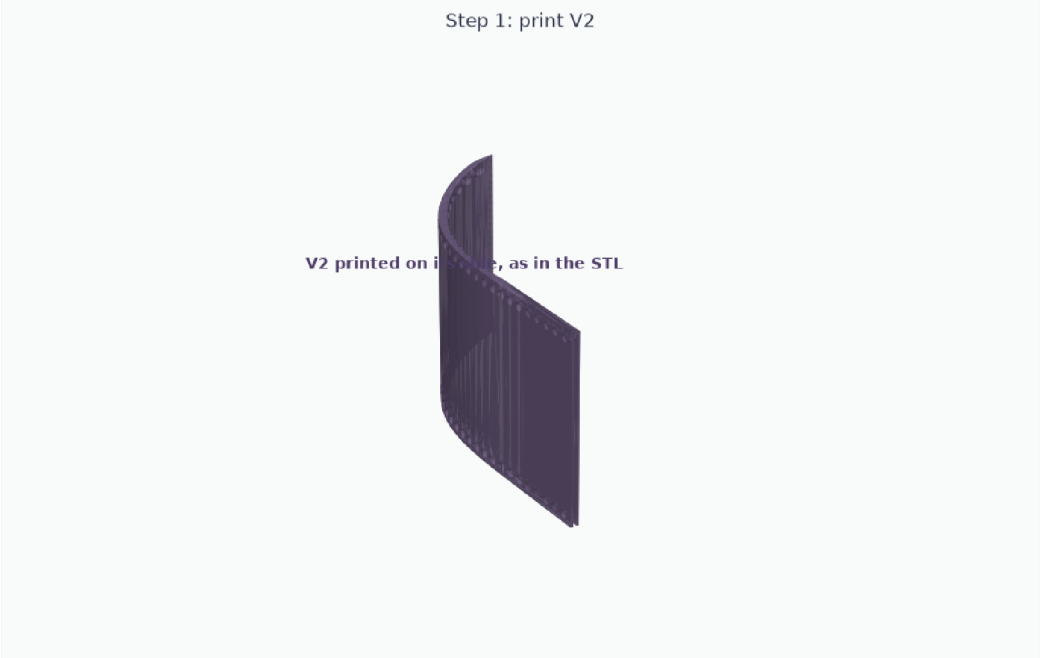
Plate 368 (5x5)

- PETG or PLA+, printed flat
- **100% infill**
- No supports
- 40 x 40 x 1.5mm

7. Step by step

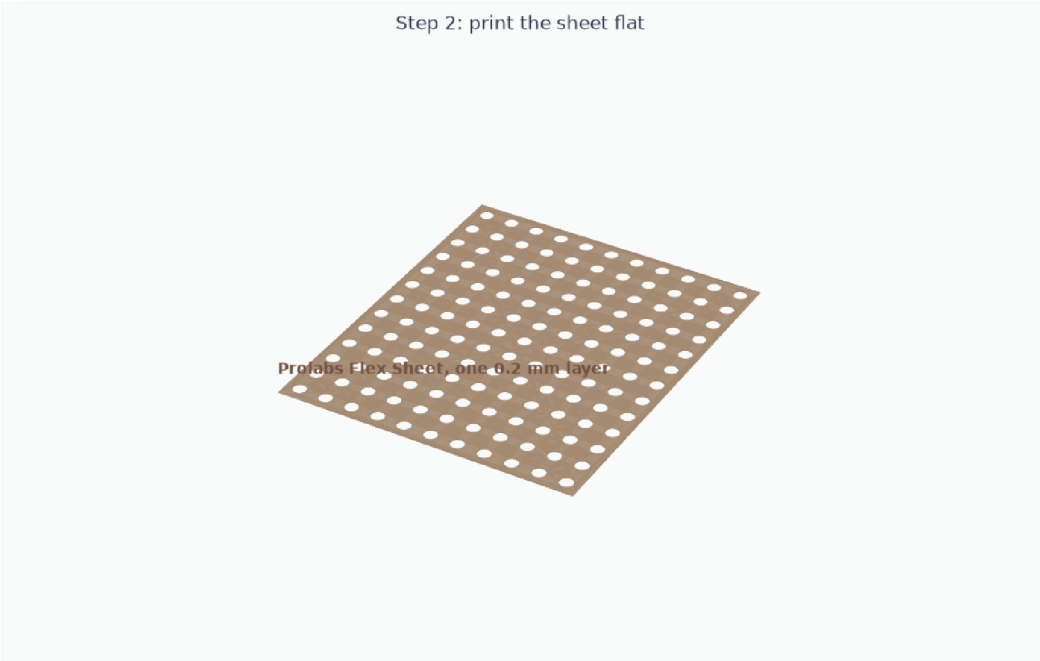
1 Print the V2 rail guard

Print **FTC Shooter V2 Rail Guard.stl** on its side. **PETG, 100% infill, no supports, 5mm brim.**



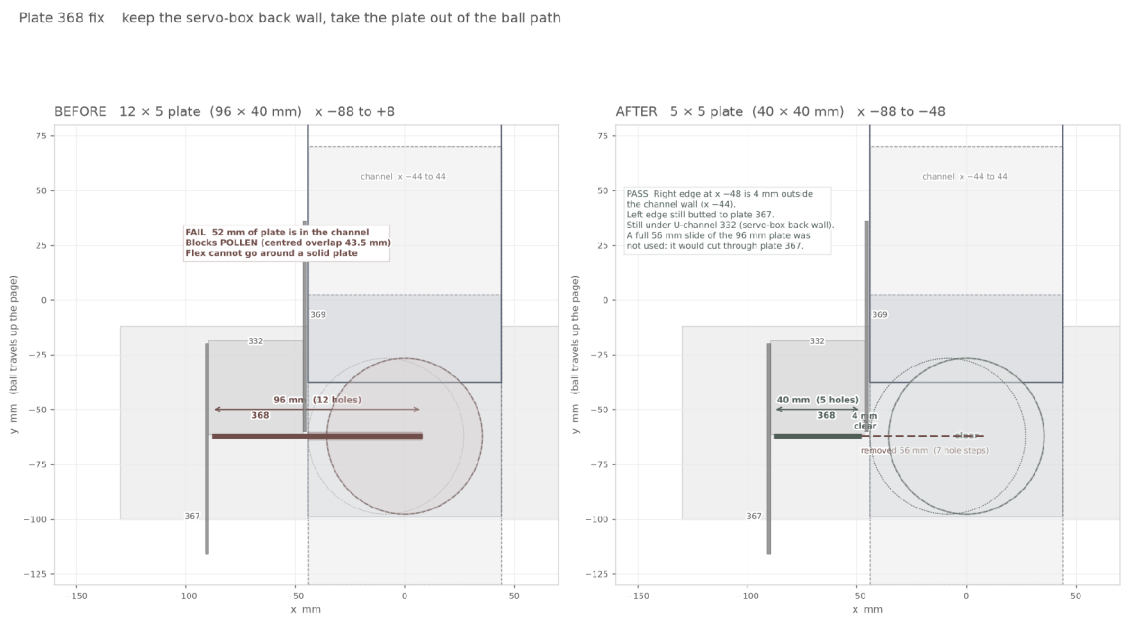
2 Print the Prolabs Flex Sheet

Print **prolabs_plain_sheet_0p2.stl** flat. It is **one 0.2mm layer**, 88 x 104mm, 143 holes of 4mm on an 8mm grid. About 1.9g.



3 Fix plate 368

Remove kit plate 368 (back wall of the left servo box, between 367 and 369, under U-channel 332). Fit **plate_368_cut_5x5.stl**, or cut 56mm (7 hole rows) off the channel end of the kit plate. **Keep the left edge against 367.** Re-use the kit screws. Check the ball has about 4mm of clearance.

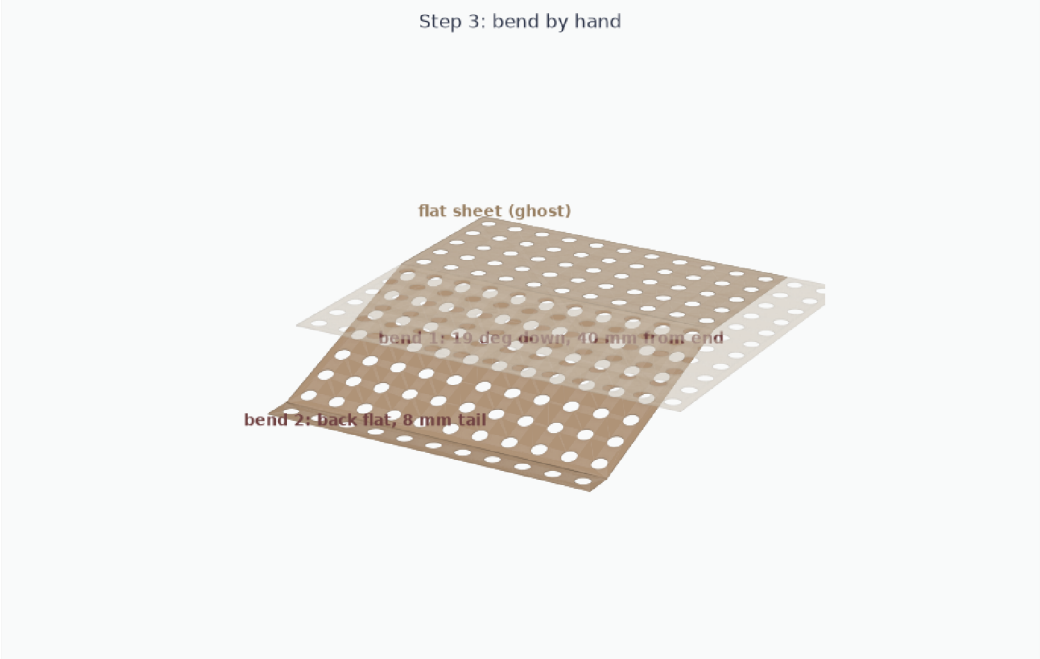


4 Mount V2 with rubber bands and zip ties

Place V2 on the shooter, between channels 316 and 317, on top of hood E. Tie it down with **at least 2 zip ties per side** (2.5mm wide ties through the 4mm holes) plus **1 rubber band per side**. The bands take up the 3 to 38mm offset between V2 holes and E holes. V2 sits 2 to 22mm above E: pull it down so it **cannot move up or down** when the ball hits it. Push on V2 by hand: no wobble.

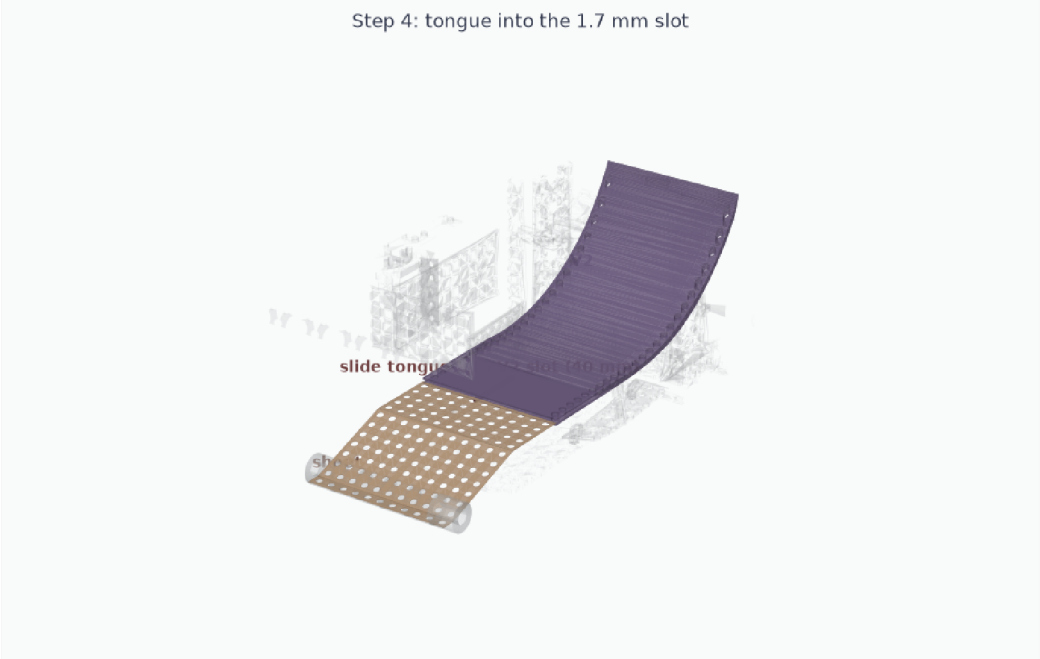
5 Bend it by hand

Bend 1: 19 deg down, 40mm (5 holes) from the tongue end. **Bend 2:** back to flat, 8mm (1 hole) from the tail end. Bend along the webs between hole rows. The sheet is thin, so it bends easily but will not hold the shape on its own.



6 Slide the tongue into the V2 slot

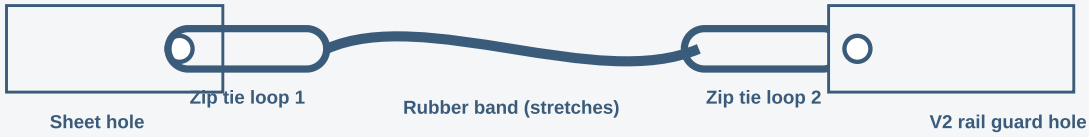
Push the 40mm tongue fully into V2's 1.7mm front slot. The tail runs down to rest on the floor plate.



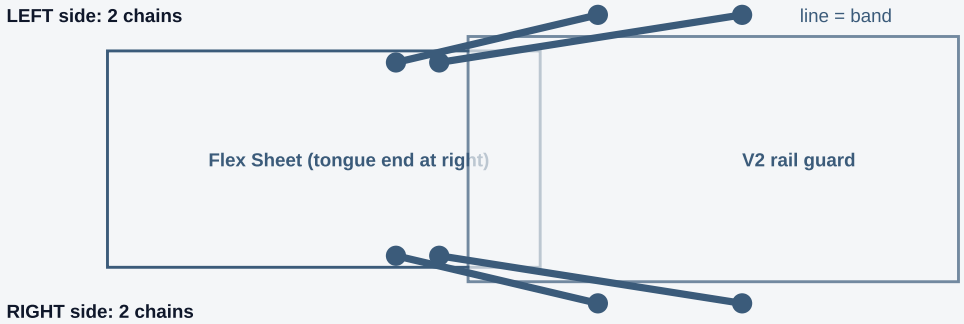
7 Fit the rubber-band chains, both sides

On **each side** make **2 chains**, 4 in total: **sheet hole → zip tie loop → rubber band → zip tie loop → V2 hole**. The bands keep the tongue pulled into the slot.

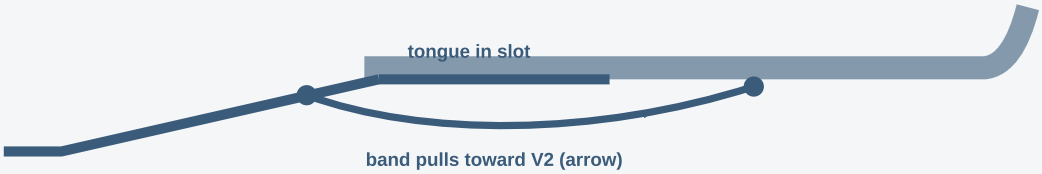
Elastic chain, top view (one chain)



Top view, whole plate: 2 chains per side

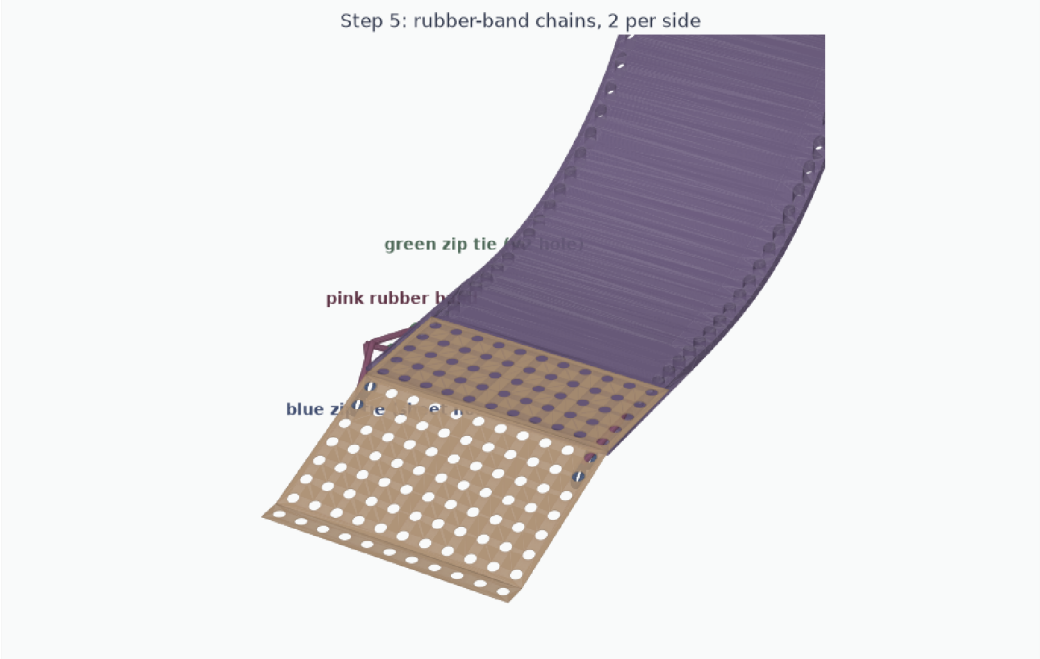


Side view (band pulls tongue into slot)



Tips

- **Do not over-tighten** the zip ties. Leave a **loose loop** so the band can stretch.
- Band tension should hold the tongue **fully seated** in the slot.
- Check that **all 4 bands match**. Replace any band that is cracked.
- V2 holes are 7.08mm apart (y -32.5, -25.4, -18.3, -11.3, -4.2) and the sheet is 8mm, so **pick holes that line up**. Use **2.5mm wide** zip ties.
- Trim zip tie tails so they do not touch the ball.



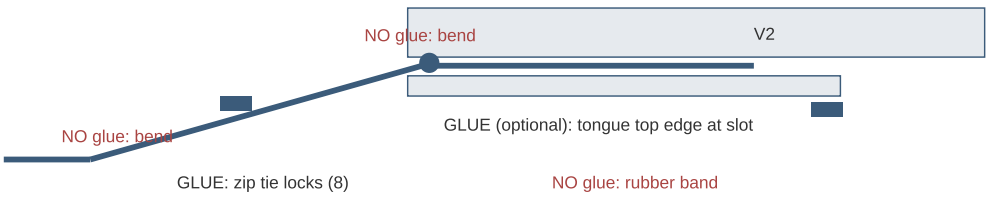
8 Super glue (thin CA)

Glue: a small drop on each of the 8 zip tie locks after tightening, so they do not slip. Optional: a thin line along the tongue's top edge where it meets the V2 slot or lip, so the tongue cannot pull out.

Do NOT glue: the two bend lines, the slope, the tail on the floor plate, or the rubber bands (CA makes rubber brittle). The bend must stay flexible.

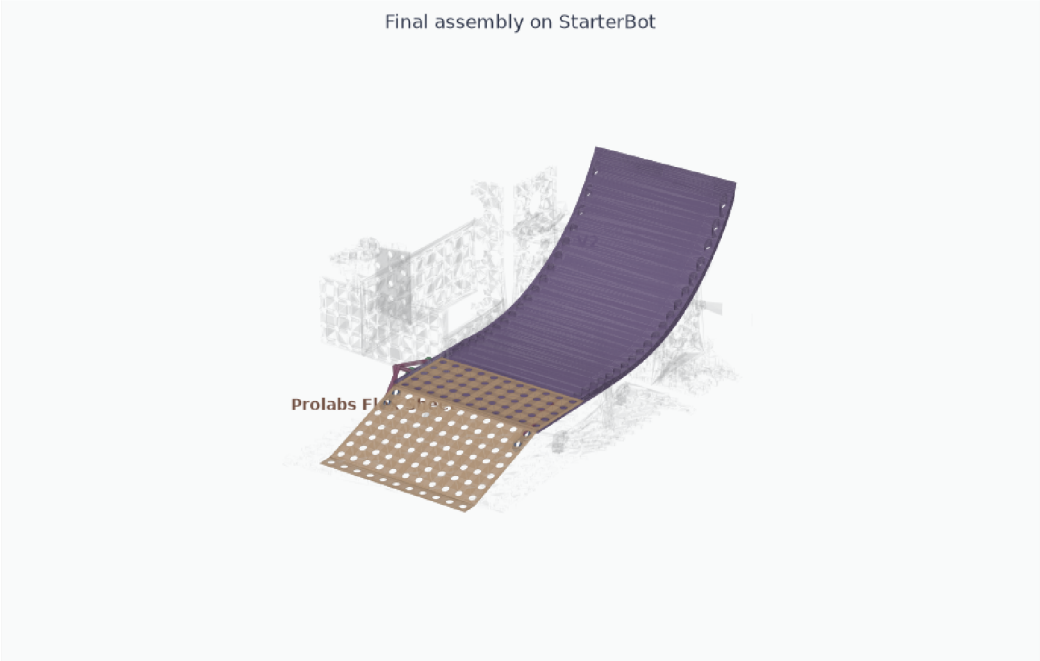
PETG note: CA sticks poorly to PETG. Lightly sand, wipe with IPA, then use a CA primer first. Cure: about 30 to 60 seconds to handle, 24 hours full strength. Do not run the robot before 1 hour.

Glue points (side view)



9 Final assembly

Check V2 is tied down (step 4). Check the tail rests on the floor plate and the sheet runs smoothly from floor to V2.



10 Hand check

Run the checklist below before using it on a robot.

8. Known issues (read before building)

1. **0.2mm sheet sags under the ball**
   A single 0.2mm layer will droop under a 71mm ball and will not hold a bend between tie points. The ball path shape is not controlled. We think a thicker sheet (0.6 to 0.8mm) or ribs would be much safer.

2. **Can tear at the holes**
   If the bands pull hard, the thin sheet can tear between holes. Use light bands and check after every run.

3. **Hole pitch 7.08mm vs 8mm**
   V2 holes at x +-40 are 7.08mm apart (under the slot: y -32.5, -25.4, -18.3, -11.3, -4.2, measured in the STL). The sheet is on 8mm, so holes are 0.1 to 2.7mm off. All 5 work with 2.5mm zip ties.

4. **1.1mm step at V2 lip**
   The sheet sits in the slot under V2's top skin, so the ball meets a 1.1mm step up onto V2. Chamfer the V2 lip edge.

5. **Ball coasts undriven**
   About 131mm with nothing driving it, partly uphill. It needs about 0.7 m/s leaving the intake roller, or a push from the next ball.

6. **Kit part 368 blocks the path (fixed)**
   The kit plate at y -62 reaches about 30mm into the ball path. Replace it with plate_368_cut_5x5.stl or cut 7 hole rows off (section 2). Then 4mm clear.

7. **NECTAR ball (91mm) does not fit**
   The channel is 88mm. Only the 71mm POLLEN ball fits, with 8.5mm per side.

8. **Not tested on a robot**
   Everything here is checked in CAD only.

9. Hand-check checklist

- [ ] V2 size within **+-0.5mm** of 129.35 x 148.57 x 88mm
- [ ] Sheet is one clean layer, no tears at holes
- [ ] Tongue goes fully 40mm into the slot
- [ ] Every zip tie lock has a glue drop and does not slip
- [ ] No glue on the two bend lines or on the band
- [ ] All 4 chains fitted, 2 per side, bands match
- [ ] Pull the sheet gently: it springs back into the slot
- [ ] Roll the 71mm ball by hand: watch how much the sheet sags
- [ ] V2 lip chamfered, no catch
- [ ] Plate 368 replaced or cut, left edge against 367, ball clears it by about 4mm
- [ ] V2 tied down with at least 2 zip ties per side, no up and down movement
- [ ] Bands stretch over the 42 to 49mm span without overstretching
- [ ] Floor tilt and the 19.5mm roller gap checked on the robot
- [ ] POLLEN balls only (NECTAR 91mm does not fit)

10. V2 + Flex Sheet vs V5b

	V2 + Prolabs Flex Sheet	V5b
Parts	3 prints (V2, sheet, plate 368) + chains	**1 print**
Material	55 cm3 (about 70g) + 1.9g sheet	227 cm3
Squeeze	14.5%	12.6%
Ramp shape control	Low (0.2mm sags, hand bent)	**Fixed shape**
Team-to-team variation	Higher	**Low**

Honest note: for untested manufacture for many teams, we think V5b is the lower-risk choice. V2 + Flex Sheet is light and cheap to try.

by Aarav Jaisingh, Prolabs Robotics. Feedback from teams is very welcome.